SUBCLINICAL hypothyroidism

(raised TSH with normal T4)

NICE 2019, NG145, BJGP 2016;66:538, Clinical Endocrinology 2016;84:799

Red Whale
GEMS
Guidelines & Evidence Made Simple

Subclinical hypothyroidism in adults (this GEMS is NOT applicable to pregnant women/trying to conceive)

There are several uncertainties surrounding the management of subclinical hypothyroidism:

- Does treatment confer long-term health benefits?
- Which subgroups of patients may benefit from treatment?

What is the evidence for treatment?

Risk of cardiovascular disease (JAMA 2010;304:1365)**:**

- In general, those with subclinical hypothyroidism have no increased risk of coronary heart disease or mortality. Those with a TSH >10 were shown to have an increased coronary heart disease risk and coronary heart disease mortality.

Impact of treatment on significant endpoints (JAMA 2018;320:1349, NEJM 2017;376:2534)**:**

- Trials assessing treatment do not show significant improvements in symptoms/quality of life.
- These studies did not assess whether treatment had any impact on cardiovascular risk/risk factors.

Management

Subclinical hypothyroidism = raised TSH with normal T4.

- Always repeat TSH test after 3m: only act if subclinical hypothyroidism confirmed on second test.
- If abnormal, check thyroid peroxidase antibodies (if present, increased risk of progression to clinical hypothyroidism).
- If >65y, be more hesitant about treating: risks from over suppression of TSH (e.g. AF) may outweigh benefits.

If TSH above normal but <10mIU/litre (on 2 tests 3m apart):

NICE says:

About half will return to normal without treatment

Consider 6m trial of treatment if <65y AND symptomatic

Stop treatment if symptoms remain once TSH within normal range

If TSH ≥10mIU/litre (on 2 tests 3m apart)

NICE says:

consider treatment with levothyroxine

(NICE says may improve symptoms/have long-term CVD benefits, but note evidence above)

NICE says 'consider' treatment in these circumstances, but what does this mean?

Consider features that may suggest underlying thyroid disease or increase the risk of progression: hypothyroidism symptoms, positive antibodies, previous radioactive iodine treatment or thyroid surgery.

- A cross-sectional study looked at common symptoms of hypo/hyperthyroidism in older adults (≥65y). 94% were euthyroid, 5% had subclinical hypothyroidism and 1% had subclinical hyperthyroidism. None had overt disease. The presence or absence of symptoms was no different between the three groups. **This suggests we need to be careful about attributing symptoms to subclinical disease** (DOI: https://doi.org/10.3399/bjgp20X708065).

If starting levothyroxine

Start levothyroxine just as you do in clinical hypothyroidism:

- Take at least 30 minutes before breakfast, caffeine and other drugs (BNF).
- If ≥65y or known cardiovascular disease, start at 25–50 micrograms/day and titrate up as required.
- For everyone else, straight to estimated treatment dose: 1.6micrograms/kg/day (to the nearest 25mcg) (for a 60kg person = 100mcg, for an 80kg person = 125mcg, for a 100kg person = 150mcg).

Risks of progression to clinical hypothyroidism if subclinical hypothyroidism untreated

2–6% of those with SUBCLINICAL disease will progress to CLINCIAL hypothyroidism each year. Risk factors for progression are being female, older and having positive thyroid peroxidase antibodies (NEJM 2017;376:2556).

If NOT treating subclinical hypothyroidism, NICE suggests checking TSH and T4 periodically:

- Features of underlying thyroid disease (e.g. raised thyroid autoantibodies, previous thyroid surgery): **annually.**
- NO features of underlying thyroid disease: **every 2–3y.**